NumPy Functions Cheat Sheet

Array Creation

Function	Description	Example
np.array()	Create array from list/tuple	np.array([1,2,3])
np.arange()	Create array with range of values	np.arange(0,10,2)
np.linspace()	Evenly spaced values between limits	np.linspace(0,1,5)
np.zeros()	Array of zeros	np.zeros((2,3))
np.ones()	Array of ones	np.ones((3,2))
np.eye()	Identity matrix	np.eye(3)
np.full()	Array with constant value	np.full((2,2),7)

Mathematical Functions

Function	Description	Example
np.add()	Element-wise addition	np.add(a,b)
np.subtract()	Element-wise subtraction	np.subtract(a,b)
np.multiply()	Element-wise multiplication	np.multiply(a,b)
np.divide()	Element-wise division	np.divide(a,b)
np.power()	Raise elements to power	np.power(a,2)
np.sqrt()	Square root	np.sqrt(a)
np.exp()	Exponential (e^x)	np.exp(a)
np.log()	Natural logarithm	np.log(a)

Statistical Functions

Function	Description	Example
np.mean()	Mean of array	np.mean(a)
np.median()	Median of array	np.median(a)
np.std()	Standard deviation	np.std(a)
np.var()	Variance	np.var(a)
np.sum()	Sum of all elements	np.sum(a)
np.min()	Minimum value	np.min(a)
np.max()	Maximum value	np.max(a)

Array Manipulation

Function	Description	Example
np.reshape()	Change shape	np.reshape(a,(2,3))
np.ravel()	Flatten array	np.ravel(a)
np.transpose()	Transpose array	np.transpose(a)
np.concatenate()	Join arrays	np.concatenate((a,b),axis=0)
np.hstack()	Horizontal stack	np.hstack((a,b))
np.vstack()	Vertical stack	np.vstack((a,b))
np.split()	Split array	np.split(a,3)

Linear Algebra

Function	Description	Example
np.dot()	Dot product	np.dot(a,b)
np.matmul()	Matrix multiplication	np.matmul(a,b)
np.linalg.inv()	Inverse of matrix	np.linalg.inv(a)
np.linalg.det()	Determinant	np.linalg.det(a)
np.trace()	Sum of diagonal elements	np.trace(a)